

Interpreter variants/extensions using monads

Luca Roversi

Exercises

- Rise "Div by 0" using the monad `Maybe` following:
 - `Arithmetic.Interpreters.Monadic.QuoRem.Exercises.EvalErrMB`
 - `Arithmetic.Interpreters.Monadic.QuoRem.Exercises.EvalErrMB_SOLUTION`
(restricted)
 - Write an interpreter that rises the exceptions:
 - "Div. by 0"
 - "Negative value error" when interpreting '`Tsub x y`' with `x` strictly smaller than `y`
- filling `Arithmetic.Interpreters.Monadic.QuoRem.Exercises.EvalSubDivErrMsg`
- Redesign an interpreters according to the following requests:
 - Consider the quotient/remainder operation on `BN` as in
`Arithmetic.Interpreters.Monadic.QuoRem.Exercises.OperationQuoRem`
 - Complete `Arithmetic.Interpreters.Monadic.QuoRem.Exercises.EvalQuoRemErrMB` to obtain an interpreter for the expressions extended as described in the previous point that rises:
 - "Div by 0" when `d` is interpreted as zero in '`Tquo n d`'
 - "Div by 0 remainder" when `d` is interpreted as zero in '`Trem n d`'
 - "Negative value" when `x` strictly smaller than `y` in '`Tsub x y`'